

Session 2 Practical

A. Playing With Numbers

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ahmed has a problem in his first year of primary school. He can not write numbers.

So his teacher suggested that he write the numbers several times, under one condition: if the teacher says "7", then Ahmed will write this number 7 times.

Write a program that will help Ahmed in doing his task.

Input

The input consists of a single line containing one integer X ($1 \leq X \leq 10$).

Output

Print the number X , repeated X times. The numbers should be separated by one space.

Sample test(s)

input
7
output
7 7 7 7 7 7 7

input
6
output
6 6 6 6 6 6

B. Factorial

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Given a number N , compute its factorial $\text{Fact}(N)$.

Where $\text{Fact}(N) = N * (N-1) * (N-2) * \dots * 3 * 2 * 1$.

Input

The input consists of one integer N ($1 \leq N \leq 20$).

Output

Print the factorial of N .

Sample test(s)

input
3
output
6

input
5
output
120

C. Digits Summation

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ismail and Omar challenged each other to a game. The game is, given a non-negative number X , they have to calculate the sum of the digits of X . The winner is the person who will say the sum first. Ismail wants to beat Omar, so he needs your help.

Input

The input consists of a number X ($0 \leq X \leq 10^{18}$).

Output

Your program should print the summation of the digits of the number X .

Sample test(s)

input
3
output
3

input
121
output
4

D. Triangle

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Mohamed has learned loops in the acmASCIS session and wants to print a triangle on the screen. Mohamed's triangle will consist of asterisks '*' only. Help him print the triangle.

Input

The input consists of one integer N ($1 \leq N \leq 100$).

Output

Print Mohamed's triangle using only asterisks.

Sample test(s)

input
3
output
* ** ***

input
6
output
* ** *** **** ***** *****

Note

- Look to sample tests for more details.

E. Calculator Again!

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Koko has attended the second session in acmASCIS level 1, and now he wants to make a more advanced calculator. His new calculator will do the same operations as the old one (addition, subtraction, multiplication and division of two integers), but it can also make multiple operations at a time, and it makes them in order.

Input

The input consists of a sequence of operations. Each operation consists of an integer X then symbol ('+' or '-' or '*' or '/', without the single quotes), then another integer Y ($0 < |X|, |Y| < 10^4$). The character '.' ends the sequence.

Output

For each operation in the sequence of operations, print the result of the calculation that was made by Koko's new calculator.

Sample test(s)

input
3+5*2.
output
16

input
12-2/2.
output
5

Note

- The sequence will contain at least one operation.
- For every operation except the first one, X represents the result of the previous operations.
- The division in this calculator is an integer division.
- The maximum number of operations in the sequence is 11.
- Make sure that the final result will be less than 2^{64} .